Explainable Quantum-Inspired Multi-label Classification of Non-Functional Requirements

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Abstract. Non-functional requirements (NFRs) frequently overlap across quality attributes such as security, performance, usability, and availability, making NFR classification a natural multi-label problem. This paper presents an intrinsically explainable quantum-inspired method for multi-label NFR classification. Requirements are represented as normalized semantic states, NFR labels as projection directions, and label scores as squared projection amplitudes; a label interference matrix models NFR co-occurrence, and token-level amplitude contributions explain predictions without a post-hoc explainer. The paper also gives a theoretical characterization of the deployed decision rule, identifying when the quantum-inspired geometry reduces to cosine-style ranking and when interference, or positive-model row-wise normalization, creates genuine cross-label coupling. We evaluate quantum-inspired variants on the NICE multi-label dataset against TF-IDF Logistic Regression, TF-IDF Linear SVM, Sentence-BERT, fine-tuned DistilBERT, and a label-frequency baseline. Under 5-fold cross-validation, Sentence-BERT achieves the highest Macro-F1; the hybrid quantum–SVM model is competitive with TF-IDF baselines and obtains the highest Macro-F1 among non-embedding models, but its advantage is not statistically significant at this dataset size. The main contribution is therefore not a state-of-the-art accuracy claim, but an interpretable multi-label scoring formulation, a precise account of when its geometry helps or collapses to classical behavior, and a deletion-based faithfulness evaluation of its intrinsic token explanations.

Keywords: Requirements Engineering · Non-Functional Requirements · Multi-label Classification · Quantum-inspired Model · Explainable AI · Software Engineering

1 Introduction

Requirements Engineering (RE) is a critical activity in software development because requirement quality strongly affects downstream architecture, implementation, testing, and maintenance decisions. Among requirement types, non-functional requirements (NFRs) are particularly difficult to identify and classify automatically. NFRs describe quality attributes such as security, performance, usability, availability, scalability, and maintainability. Unlike many functional requirements, NFRs are often ambiguous, context-dependent, and overlapping.

For example, a requirement stating that payment information must be encrypted and accessible only to authorized administrators clearly relates to security. If the same requirement also constrains availability, auditability, or response time, it may express multiple NFR categories at once. Therefore, NFR classification is better treated as a multi-label problem rather than a strict single-label problem [19,21].

This paper investigates the research topic: *Explainable Quantum-Inspired Multi-label Classification of Non-Functional Requirements*. The key idea is to represent each requirement as a semantic state and estimate its degree of association with multiple NFR labels using projection amplitudes. The approach is quantum-inspired rather than quantum-hardware-based: it runs on ordinary computers and borrows mathematical intuitions such as state representation, projection amplitude, and interference from geometric information retrieval and quantum language modeling [16,18,8].

The paper's positive claim is that this formulation provides an intrinsic explanation mechanism for multi-label NFR classification and a testable theory of when quantum-inspired geometry changes the decision rule. The basic projection component is closely related to centroid-based text classification; the interference and positive-model row-wise normalization analyses make explicit when the model becomes more than an independent centroid classifier and when it does not. Empirically, the hybrid model offers a small, suggestive accuracy gain over comparable TF-IDF baselines, while the primary value of the method lies in its transparent scoring structure and diagnostic ablation.

The contributions of this paper are:

- an intrinsically explainable quantum-inspired formulation for multi-label NFR classification;
- a theoretical characterization of interference and, for the positive-projection model, row-wise normalization as two sources of cross-label coupling;
- token-level semantic amplitude explanations evaluated with a deletion-based ERASER faithfulness test;
- empirical comparison against TF-IDF Logistic Regression, TF-IDF Linear SVM, Sentence-BERT, fine-tuned DistilBERT, and a label-frequency baseline on the NICE dataset;
- an ablation study showing where projection type, label interference, and hybrid fusion weight help or fail.

2 Related Work

Automated NFR classification includes early rule and supervised-learning work on security, performance, usability, and related quality attributes [1,7,2]. PROMISE and PROMISE-expanded support repeatable NFR experiments [11], while NICE provides the multi-label NFR setting used here [13,14]. Because label filtering, splits, and metrics differ from the NICE paper, we report all baselines on the same downloaded data.

Our baselines cover TF-IDF classifiers, which remain strong in small text datasets [17], and transformer-based representations, including BERT-style transfer learning, NoRBERT, and Sentence-BERT [4,6,12]. The quantum-inspired component follows geometric information retrieval and quantum-inspired language modeling, where matching is expressed through vector spaces, projections, and wave-function-inspired representations [16,18,22,8]. For explainability, we relate to LIME, SHAP, and ERASER-style rationale evaluation [15,10,5].

3 Research Questions

This study is guided by three research questions:

RQ1. Can a quantum-inspired semantic projection model classify multi-label NFRs competitively compared with classical and embedding baselines?

RQ2. Does the model provide useful explainability through token-level semantic amplitude contributions?

RQ3. Is the observed behavior stable under cross-validation rather than only in a single train/test split?

4 Proposed Method

4.1 Requirement Representation

Each requirement text is first converted into a TF-IDF vector:

$$\mathbf{x}_i \in \mathbb{R}^d, \tag{1}$$

where d is the vocabulary size. The vector is normalized to form a semantic state:

$$|r_i\rangle = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2}. \tag{2}$$

This normalization allows the requirement to be interpreted as a direction in semantic space.

4.2 Label Projection

For each NFR label c, a label basis vector is estimated from the centroid of positive training examples:

$$|c\rangle = \frac{1}{N_c} \sum_{i:y_{ic}=1} |r_i\rangle, \tag{3}$$

followed by normalization. The raw association between requirement r_i and label c is computed as a squared projection amplitude:

$$s_c(r_i) = \left(\langle c \mid r_i\rangle\right)^2. \tag{4}$$

We call $s_c(r_i)$ an association score rather than a probability because the scores over labels are not constrained to sum to one.

4.3 Contrastive Label Projection

The positive-only centroid projection is simple and interpretable, but it does not explicitly model how a label differs from other labels. Therefore, this work also evaluates a contrastive quantum-inspired variant. For each label c, the contrastive direction is computed as:

$$|c_{\Delta}\rangle = \frac{\mu_c^+ - \mu_c^-}{\left\|\mu_c^+ - \mu_c^-\right\|_2}, \tag{5}$$

where μ_c^+ is the centroid of requirements that contain label c, and μ_c^- is the centroid of requirements that do not contain label c. Unlike the positive centroid vector, the contrastive direction is signed: a requirement can project in the opposite direction of a label. Squaring a negative projection would incorrectly assign a high score to evidence against the label. Therefore, the contrastive variant uses a half-wave rectified projection score:

$$s_c^{\Delta}(r_i) = \left(\max(0, \langle c_{\Delta} \mid r_i\rangle)\right)^2. \tag{6}$$

This preserves positive alignment with the label direction while suppressing opposite-direction evidence.

4.4 Interference Adjustment

To reflect label co-occurrence, the model estimates a label interference matrix from training labels. Let N_{kl} be the number of training samples containing both labels k and l, and let N_k be the number of samples containing label k. The normalized co-occurrence matrix is:

$$M_{kl} = \frac{N_{kl}}{N_k}, \tag{7}$$

with the diagonal set to 1. The implementation mixes this matrix with the identity matrix:

$$\mathbf{M}' = (1 - \lambda)\mathbf{I} + \lambda\mathbf{M}, \tag{8}$$

where λ controls interference strength. For projection scores $\mathbf{p}_i$, adjusted scores are computed as:

$$\mathbf{s}_i = \text{normalize}(\mathbf{p}_i\mathbf{M}'). \tag{9}$$

For the positive-projection model, normalize$(\cdot)$ denotes row-wise min–max normalization into $[0, 1]$. For the contrastive model, the rectified projection is passed through a sigmoid calibration and the interference-adjusted scores are clipped into $[0, 1]$. The resulting scores are association scores for thresholding rather than mutually exclusive probabilities.

4.5 Hybrid Quantum–SVM Fusion

To improve stability on small and imbalanced datasets, a hybrid model combines the contrastive quantum-inspired score with a one-vs-rest Linear SVM score:

$$\mathbf{h}_i = \alpha \mathbf{q}_i + (1 - \alpha)\mathbf{v}_i, \tag{10}$$

where $\mathbf{q}_i$ is the contrastive quantum-inspired score vector, $\mathbf{v}_i$ is the calibrated SVM score vector, and α is the quantum fusion weight. In the final experiments, $\alpha = 0.30$ is used as a moderate quantum weight; the ablation study (Table 5) shows that Macro-F1 is not sensitive to the exact value in this range ($\alpha = 0.15$, 0.30, and 0.50 are within one standard deviation of each other), so $\alpha = 0.30$ was kept as a round, non-edge value rather than as a value selected for maximizing cross-validation Macro-F1.

4.6 Explainability

For a predicted label c, token-level contribution is estimated from the element-wise product between the requirement state and the label basis:

$$e_{ijc} = r_{ij} c_j. \tag{11}$$

Terms with the largest absolute contributions are reported as explanations. In the experiments, these explanations are generated by the rectified contrastive projection model. We evaluate them using a deletion test: if the highlighted terms are faithful, deleting them should reduce the model score more than deleting random nonzero terms.

4.7 Theoretical Characterization

The positive-projection formulation exposes two mechanistically distinct sources of cross-label coupling in its deployed decision rule: the interference matrix (Proposition 1) and the row-wise score normalization applied by this variant before thresholding (Proposition 2). Only the within-instance *ranking* of labels reduces to a cosine-similarity ranking; the thresholded binary decision of this variant does not reduce to L independent per-label centroid classifiers, even when interference is switched off. The contrastive and hybrid variants instead use sigmoid-calibrated scores without row-wise min–max normalization, so Proposition 2 does not apply to them.

Proposition 1 (Interference as label coupling). *With* $\mathbf{M}' = (1-\lambda)\mathbf{I} + \lambda\mathbf{M}$, *the pre-normalization score for label* l *has the form*

$$u_l \propto (1 - \lambda)p_l + \lambda \sum_k p_k M_{kl}. \tag{12}$$

The second term injects score mass from co-occurring labels k, *where* $M_{kl} = N_{kl}/N_k$. *This coupling cannot be represented by a diagonal per-label centroid* *classifier; together with the row-wise normalization in Proposition 2, it is one of two mechanisms by which the positive-projection decision rule departs from an independent per-label classifier.*

Proof. Expanding $\mathbf{pM}'$ gives $\mathbf{p}((1-\lambda)\mathbf{I}+\lambda\mathbf{M})$, so the l-th component is $(1-\lambda)p_l+\lambda\sum_k p_k M_{kl}$. If $\mathbf{M}'$ were diagonal, each adjusted label score would depend only on its own projection score p_l. The off-diagonal entries M_{kl} for $k \neq l$ therefore introduce explicit cross-label coupling through co-occurrence statistics.

Proposition 2 (Ranking invariance under row-wise normalization). *Let $\mathbf{u}_i = \mathbf{p}_i\mathbf{M}'$ be the pre-normalization score vector from Proposition 1, and let $\mathbf{s}_i = \text{normalize}(\mathbf{u}_i)$ (row-wise min–max into $[0,1]$) be the score vector actually thresholded by the implementation. For any two labels k, l evaluated on the same instance i,*

$$s_{i,k} \geq s_{i,l} \iff u_{i,k} \geq u_{i,l};$$

in particular, at $\lambda = 0$ (so $u_{i,l} = p_{i,l} = \langle l \mid r_i \rangle^2$), this ranking equals the ranking by cosine similarity $\langle l \mid r_i \rangle$. However, for a fixed global threshold $\tau \in (0,1)$, the decision $s_{i,l} \geq \tau$ is equivalent to

$$u_{i,l} \ \geq \ \min_k u_{i,k} + \tau\Big(\max_k u_{i,k} - \min_k u_{i,k}\Big) =: \tau_i,$$

where τ_i depends on instance i through the scores of every *label on that instance, not only label l. Consequently, τ does not correspond to a single instance-independent cutoff on $u_{i,l}$ (or, at $\lambda = 0$, on cosine similarity): two instances with an identical raw score for label l can receive opposite decisions for that label if their scores on other labels differ.*

Proof. For fixed i with $\max_k u_{i,k} > \min_k u_{i,k}$, $s_{i,l} = (u_{i,l} - \min_k u_{i,k})/(\max_k u_{i,k} - \min_k u_{i,k})$ is a strictly increasing affine function of $u_{i,l}$ with the same slope and intercept for every label l of instance i, which proves the ranking-invariance claim. At $\lambda = 0$, $u_{i,l} = \langle l \mid r_i \rangle^2$, and since $\langle l \mid r_i \rangle = \cos\theta_{r_i,l} \geq 0$ (TF-IDF features and positive centroids are non-negative), squaring is order-preserving on $[0,\infty)$, so the ranking by $u_{i,l}$ equals the ranking by $\cos\theta_{r_i,l}$. Solving $s_{i,l} \geq \tau$ for $u_{i,l}$ using the definition of $s_{i,l}$ gives the threshold τ_i above, which is a function of $\min_k u_{i,k}$ and $\max_k u_{i,k}$ and therefore of the scores of labels $k \neq l$ on instance i; two instances with equal $u_{i,l}$ but different label-score profiles can therefore yield different values of τ_i relative to $u_{i,l}$, hence different decisions. When $\max_k u_{i,k} = \min_k u_{i,k}$, the implementation sets $s_{i,l} = 0$ for all l, so no label is predicted for instance i at any $\tau > 0$.

Remark 1. For two labels and $\tau = 0.5$, $\mathbf{u} = (0.5, 0.0)$ predicts label 1 after normalization, whereas $\mathbf{u} = (0.5, 0.9)$ gives the same raw label-1 score but does not predict label 1. Row-wise normalization is therefore a second source of cross-label coupling, independent of $\mathbf{M}'$.

5 Experimental Setup

5.1 Datasets

NICE multi-label dataset. The main experiment uses the NICE dataset from Zenodo [13]. The downloaded CSV contains 622 requirements and binary label columns for NFR classes. After filtering rows with at least one NFR sub-class label and removing the empty "Other" label, the experiment uses 381 requirements and 11 NFR labels. Among these, 125 requirements contain more than one NFR label, giving a label cardinality of 1.3648.

PROMISE-expanded dataset. A secondary experiment uses PROMISE-expanded from the public requirements classification repository [11]. This dataset is treated as an 11-class single-label NFR subtype classification task and is used as an auxiliary sanity check rather than the main multi-label experiment.

5.2 Compared Models

The compared baselines are:

- **Label-frequency baseline:** predicts labels according to training label frequencies.
- **TF-IDF Logistic Regression:** one-vs-rest Logistic Regression using unigram and bigram TF-IDF features.
- **TF-IDF Linear SVM:** one-vs-rest Linear SVM using unigram and bigram TF-IDF features.
- **Sentence-BERT + LR:** all-MiniLM-L6-v2 embeddings followed by one-vs-rest Logistic Regression.
- **Fine-tuned DistilBERT:** distilbert-base-uncased fine-tuned directly for multi-label classification with BCE loss and per-label threshold calibration.

The quantum-inspired models are:

- **Quantum-inspired Projection:** positive-centroid semantic projection with label interference.
- **Contrastive Quantum Projection:** positive-minus-negative semantic projection with label interference.
- **Hybrid Quantum–SVM Fusion:** score-level fusion of contrastive quantum projection and Linear SVM, using a quantum fusion weight of 0.30.

5.3 Metrics and Threshold Selection

The main reported metrics are Micro-F1, Macro-F1, Hamming loss, and Label Ranking Average Precision (LRAP). Macro-F1 is emphasized because the NFR labels are imbalanced. Two threshold protocols are evaluated. First, a single global threshold is selected on an internal validation split by maximizing Macro-F1 over thresholds from 0.05 to 0.95. Second, a per-label threshold protocol selects an independent threshold for each NFR label on the validation split. Fine-tuned DistilBERT was evaluated only with the per-label protocol; no global-threshold DistilBERT result is reported.

5.4 Hyperparameters

Table 1 summarizes the main hyperparameters used for reproducibility.

Table 1. Main hyperparameters used in the experiments.

Component	Setting
TF-IDF features	unigrams+bigrams, min_df=1
Sublinear TF-IDF	enabled for contrastive, hybrid, and SVM-only ablation
Logistic Regression	one-vs-rest, class_weight=balanced, max_iter=3000
Linear SVM	one-vs-rest LinearSVC, class_weight=balanced
Sentence-BERT	all-MiniLM-L6-v2 + one-vs-rest Logistic Regression
Fine-tuned DistilBERT	2 epochs, batch size 16, max length 128, learning rate 5×10^{-5}
Positive projection interference	$\lambda = 0.15$
Contrastive projection interference	$\lambda = 0.05$
Hybrid projection interference	$\lambda = 0.03$
Hybrid quantum weight	$\alpha = 0.30$ unless varied in ablation
Threshold grid	0.05 to 0.95 with step 0.05
Random seed	42

6 Results

6.1 Single Train/Validation/Test Split on NICE

Table 2 reports results from a single train/validation/test split on the NICE dataset.

Table 2. NICE multi-label results on a single train/validation/test split, comparing label-frequency, TF-IDF, Sentence-BERT, and quantum-inspired models across Micro-F1, Macro-F1, Hamming loss, and LRAP. Sentence-BERT has the highest Macro-F1 and LRAP on this split; the hybrid quantum–SVM model and the contrastive quantum projection share the lowest Hamming loss.

Model	Threshold	Micro-F1	Macro-F1	Hamming Loss	LRAP
Label-frequency baseline	0.05	0.2221	0.2166	0.8751	0.4183
TF-IDF Logistic Regression	0.45	0.6299	0.5266	0.0901	0.7861
TF-IDF Linear SVM	0.45	0.6212	0.5205	0.0877	0.7843
Sentence-BERT + LR	0.50	**0.6667**	**0.6049**	0.0901	**0.8286**
Quantum-inspired Projection	0.85	0.6076	0.5618	0.0980	0.7694
Contrastive Quantum Projection	0.50	0.4907	0.3885	**0.0870**	0.6818
Hybrid Quantum–SVM Fusion	0.45	0.6284	0.5286	**0.0870**	0.7779

On this single split, Sentence-BERT obtains the highest Macro-F1 and LRAP. The positive projection variant performs better than the rectified contrastive variant, while the hybrid quantum–SVM model and the contrastive quantum projection share the lowest Hamming loss on this split. Because the split contains only 381 usable instances, these single-split results should be interpreted cautiously.

6.2 5-Fold Cross-validation on NICE

Table 3 reports 5-fold cross-validation results with mean and standard deviation across folds under the global-threshold protocol.

Table 3. Mean and standard deviation of 5-fold cross-validation results on the NICE multi-label dataset using one validation-selected global threshold per fold. Sentence-BERT achieves the highest LRAP; the hybrid quantum–SVM model obtains the highest Micro-F1 and Macro-F1 among non-embedding models, while TF-IDF Logistic Regression obtains the lowest Hamming loss.

Model	Micro-F1	Macro-F1	Hamming Loss	LRAP
Sentence-BERT + LR	0.6493 ± 0.0357	**0.6121 ± 0.0450**	0.0919 ± 0.0200	**0.8361 ± 0.0225**
Hybrid Quantum–SVM Fusion	**0.6839 ± 0.0331**	0.6084 ± 0.0294	0.0768 ± 0.0074	0.8040 ± 0.0266
TF-IDF Linear SVM	0.6821 ± 0.0405	0.6049 ± 0.0336	0.0764 ± 0.0094	0.8056 ± 0.0262
TF-IDF Logistic Regression	0.6787 ± 0.0463	0.5977 ± 0.0229	**0.0759 ± 0.0140**	0.7990 ± 0.0240
Contrastive Quantum Projection	0.4300 ± 0.0783	0.4118 ± 0.0275	0.2074 ± 0.1634	0.6968 ± 0.0503
Quantum-inspired Projection	0.5931 ± 0.0166	0.5490 ± 0.0237	0.1096 ± 0.0213	0.7830 ± 0.0304
Label-frequency baseline	0.2328 ± 0.0034	0.2083 ± 0.0032	0.7927 ± 0.0019	0.4059 ± 0.0110

Under standard 5-fold cross-validation with a global threshold, Sentence-BERT is the strongest Macro-F1 and LRAP baseline. The hybrid quantum–SVM fusion model is the strongest non-embedding model in Micro-F1 and Macro-F1 and improves over both pure quantum-inspired variants. A Wilcoxon signed-rank test over fold-level Macro-F1 finds no significant difference between the hybrid model and TF-IDF Linear SVM ($p = 0.1875$) or TF-IDF Logistic Regression ($p = 0.4375$). Given the small number of folds and the dependence induced by cross-validation resampling, these tests should be interpreted as descriptive evidence rather than a strong statistical conclusion [3].

6.3 5-Fold Cross-validation with Per-label Thresholds

Table 4 reports the 5-fold results when each label receives its own threshold selected on the validation split.

With per-label threshold calibration, Sentence-BERT again obtains the best Macro-F1, and the hybrid quantum–SVM model remains the strongest non-embedding model in Macro-F1. Among the models evaluated under both protocols, absolute Macro-F1 decreases relative to the global-threshold protocol (Table 3), likely reflecting the smaller per-label validation subsets used to select each threshold; the hybrid model's relative drop (-3.8%) is smaller than TF-IDF Linear SVM's (-6.8%) or TF-IDF Logistic Regression's (-5.8%). The fine-tuned DistilBERT result cannot be compared across threshold protocols because no

Table 4. Mean and standard deviation of 5-fold results with per-label threshold calibration on the NICE multi-label dataset. This is the only threshold protocol run for fine-tuned DistilBERT. Sentence-BERT has the highest Macro-F1 and LRAP, while the hybrid quantum–SVM model remains the strongest non-embedding model in Macro-F1.

Model	Micro-F1	Macro-F1	Hamming Loss	LRAP
Sentence-BERT + LR	**0.6386 ± 0.0280**	**0.5990 ± 0.0276**	**0.0947 ± 0.0092**	**0.8361 ± 0.0225**
Hybrid Quantum–SVM Fusion	0.6178 ± 0.0749	0.5854 ± 0.0341	0.1027 ± 0.0476	0.8040 ± 0.0266
TF-IDF Linear SVM	0.6085 ± 0.0588	0.5640 ± 0.0309	0.1057 ± 0.0378	0.8056 ± 0.0262
TF-IDF Logistic Regression	0.6219 ± 0.0588	0.5629 ± 0.0532	0.0967 ± 0.0195	0.7990 ± 0.0240
Fine-tuned DistilBERT	0.5420 ± 0.0569	0.5333 ± 0.0591	0.1402 ± 0.0361	0.7105 ± 0.0516

global-threshold run is available. Under the per-label protocol, it is weaker than the frozen Sentence-BERT+LR baseline and the TF-IDF/hybrid models in this small, imbalanced setting, showing that direct fine-tuning of a general-purpose encoder is not automatically superior to strong linear baselines under the present data budget.

6.4 Ablation Study

Table 5 reports a 5-fold ablation study of the main quantum-inspired components. The ablation compares the original positive projection, contrastive projection with and without interference, an SVM-only component using the same sublinear TF-IDF representation as the hybrid model, and hybrid fusion with different quantum weights.

Table 5. Ablation study on the NICE multi-label dataset. Hybrid fusion with $\alpha \in \{0.15, 0.30, 0.50\}$ performs within one standard deviation across all four metrics, indicating that the projection score is useful as a calibrated auxiliary signal but that Macro-F1 is not sensitive to the exact quantum weight in this range; $\alpha = 0.30$ is reported as the main configuration.

Variant	Micro-F1	Macro-F1	Hamming Loss	LRAP
Hybrid $\alpha = 0.15$	**0.6860 ± 0.0312**	**0.6091 ± 0.0293**	**0.0756 ± 0.0072**	**0.8085 ± 0.0280**
Hybrid $\alpha = 0.30$	0.6839 ± 0.0331	0.6084 ± 0.0294	0.0768 ± 0.0074	0.8040 ± 0.0266
Hybrid $\alpha = 0.50$	0.6814 ± 0.0404	0.6054 ± 0.0299	0.0795 ± 0.0113	0.8012 ± 0.0232
SVM-only sublinear TF-IDF	0.6713 ± 0.0553	0.5945 ± 0.0409	0.0843 ± 0.0263	0.8081 ± 0.0291
Positive projection without interference	0.5948 ± 0.0238	0.5526 ± 0.0285	0.1108 ± 0.0233	0.7839 ± 0.0309
Positive projection with interference	0.5931 ± 0.0166	0.5490 ± 0.0237	0.1096 ± 0.0213	0.7830 ± 0.0304
Contrastive projection without interference	0.5447 ± 0.0420	0.5397 ± 0.0398	0.1339 ± 0.0100	0.7183 ± 0.0523
Contrastive projection with interference	0.4300 ± 0.0783	0.4118 ± 0.0275	0.2074 ± 0.1634	0.6968 ± 0.0503

The ablation supports four observations. First, interference provides no measurable benefit for either projection variant on this dataset: it leaves the positive projection essentially unchanged (Macro-F1 0.5526 without interference versus 0.5490 with it) and substantially worsens the contrastive projection (0.5397 without versus 0.4118 with). The best hybrid configuration also uses a small interference weight ($\lambda = 0.03$), and the fusion weight that matters is α, not λ. We therefore treat interference as a secondary, largely inactive component in the current TF-IDF setting rather than as a primary contribution, and report it mainly because the underlying label-coupling formulation in Proposition 1 is independently interesting. Second, the rectified contrastive projection is semantically safer than squaring signed negative evidence, but in this dataset the pure contrastive variant performs worse than the positive projection. Third, the hybrid model improves over pure projection variants, showing the value of discriminative calibration. Compared with the SVM-only sublinear TF-IDF component, Hybrid $\alpha = 0.30$ improves mean Macro-F1 from 0.5945 to 0.6084, an absolute gain of 0.0139 attributable to adding the projection score as an auxiliary signal. Fourth, performance is best when the projection score is used as an auxiliary signal rather than as the dominant score.

6.5 Explainability Evaluation

We evaluate explanation faithfulness using the ERASER framework metrics: Comprehensiveness and Sufficiency. Across 158 true label assignments in the test split, for the contrastive projection model: (i) Comprehensiveness (score drop when deleting top-3 terms) was 0.0305, which is 5.21 times higher than random deletion (0.0059). (ii) Sufficiency is computed as $f(x)_c - f(x_{\text{top-3}})_c$, so lower values are better because the rationale-only input should preserve the original score. The contrastive projection obtains -0.0395, whereas keeping 3 random terms drops the score by 0.0337 (ratio -1.17). This negative value means the top-3 rationale alone receives a slightly higher score than the full requirement; we interpret it cautiously as evidence that non-rationale tokens can add background noise, not as a human-rationale quality guarantee. For the TF-IDF Linear SVM baseline: (i) Comprehensiveness was 0.0867 (4.01 times random drop of 0.0216). (ii) Sufficiency was 0.0484 (0.42 times random drop of 0.1141). Relative to a random-deletion control, the intrinsic quantum-inspired explanations are meaningfully faithful and of a similar order to the SVM coefficient baseline; in absolute score-drop terms, the SVM explanation removes more score mass (0.0867 vs. 0.0305), so this comparison should be read as competitive on a relative, random-normalized basis rather than as uniformly stronger.

Table 6 gives a qualitative example from the same deletion test. The highest-contribution terms for the Security label correspond to access control concepts that are also plausible human rationales.

Table 6. Example requirement with top token-level contributions for the true NFR label, illustrating the intrinsic explanation produced by the projection model.

Requirement excerpt	Label	Top terms
"The product shall ensure that it can only be accessed by authorized users"	Security	authorized, only, access

6.6 Auxiliary PROMISE-expanded Result

The auxiliary single-label PROMISE-expanded experiment suggests that the quantum-inspired projection model can be competitive in a single-label setting: on one 70/30 train/test split, it obtains 0.7025 accuracy and 0.6748 Macro-F1, compared with 0.6899 accuracy and 0.6344 Macro-F1 for TF-IDF Logistic Regression. This result comes from a single split without cross-validation or a significance test, so, like the single-split NICE result above, it should be interpreted cautiously; it is not the main evidence for the multi-label research claim, and because PROMISE-expanded is single-label, it does not exercise the multi-label interference mechanism.

7 Discussion

The experiments show a useful separation between accuracy and explanation. On the main NICE multi-label dataset, Sentence-BERT is the strongest baseline in Macro-F1 and LRAP, while the hybrid quantum–SVM model is competitive with TF-IDF baselines and provides intrinsic token-level contributions. The pure quantum-inspired projections are not competitive classifiers by themselves: they lose to TF-IDF Linear SVM in all 5 cross-validation folds, with large paired effect sizes (Cohen's $d_z = -1.36$ for positive projection and -5.92 for contrastive projection; Wilcoxon $p = 0.0625$, the smallest value attainable at $n = 5$ folds).

Hybrid fusion is the component that makes the approach practically usable. Its Macro-F1 is directionally higher than an SVM-only baseline sharing the same TF-IDF representation (+0.0139, paired $d_z = 0.49$), but the gain is not statistically reliable: it is negative in 2 of 5 folds, and all main Wilcoxon comparisons in Tables 3 and 4 are non-significant ($p \geq 0.1875$). Detecting an effect of this size in an idealized two-sided paired t-test ($\alpha = 0.05$, power $= 0.80$) would require 34 independent pairs. Cross-validation folds from one dataset are dependent, so this calculation is a sensitivity diagnostic, not a proposal to create 34 folds from the same 381 instances; hence the accuracy gain should be read as suggestive, not confirmed.

To reduce reliance on fold-level significance testing, we also ran a paired bootstrap over the held-out NICE test requirements. The hybrid model's Macro-F1 advantage over TF-IDF Linear SVM was 0.0081 with a 95% bootstrap interval of [-0.0100, 0.0263], while its advantage over TF-IDF Logistic Regression was 0.0020 with interval [-0.0205, 0.0227]. Both intervals overlap zero, corroborating the cross-validation result: at the current dataset size, the data cannot distinguish the hybrid model's accuracy from that of TF-IDF baselines. The one accuracy comparison that is both large and well-supported is not between the hybrid model and classical baselines, but between the hybrid model and the pure quantum-inspired projection it fuses (+0.140 Macro-F1, 95% CI $[0.074, 0.213]$, $p < 0.001$): the quantum projection is too weak to deploy on its own and depends on the SVM component to be usable, rather than the reverse.

Although Sentence-BERT obtains the highest Macro-F1 and LRAP, the hybrid projection model occupies a different point in the tradeoff between accuracy and interpretability. It does not add a confirmed accuracy advantage over classical baselines. Sentence-BERT relies on a pretrained transformer encoder and produces dense embeddings whose individual dimensions are not directly interpretable; explanations therefore require post-hoc methods such as LIME [15] or SHAP [10]. In contrast, the projection model yields token-level contributions intrinsic to the scoring function and passes a deletion-based faithfulness check, while remaining statistically indistinguishable from TF-IDF baselines. The method is therefore most attractive when intrinsic interpretability and a small dependency footprint matter more than maximum Macro-F1, not as an accuracy improvement in its own right.

8 Threats to Validity

Dataset size and availability. NICE contains 381 usable multi-label NFR requirements after filtering, so all cross-validation and ablation results come from one small corpus. The observed ranking may depend on NICE's vocabulary, label distribution, and annotation conventions, and a second multi-label NFR corpus is needed before making broad accuracy-generalization claims. Multi-label NFR datasets with fine-grained subclasses are scarce; as a partial mitigation, we include a single-label PROMISE-expanded sanity check, but it does not exercise the multi-label interference mechanism. The current evidence is sufficient for the paper's interpretability and diagnostic claims, but not for a state-of-the-art accuracy claim beyond NICE.

Baseline scope. The current implementation includes TF-IDF baselines, a frozen Sentence-BERT encoder, and a cross-validated fine-tuned DistilBERT baseline. It does not include NoRBERT, RoBERTa, ML-kNN, or the small-language-model classifier from the NICE paper. The DistilBERT result reduces the risk that the comparison relies only on frozen embeddings, but it is not an exhaustive neural baseline study: different pretrained encoders, longer schedules, or domain-specific requirements models could change the ranking. This does not affect the paper's central claims, which are explicitly framed around interpretability and diagnostic ablation rather than state-of-the-art accuracy.

Model simplicity. The quantum-inspired models are based on TF-IDF semantic states, projection directions, and score fusion. They do not yet use contextual neural embeddings inside the projection component.

Threshold sensitivity. Multi-label classification depends strongly on decision thresholds. The experiments show that per-label threshold calibration can change the relative ranking of models.

Explainability evaluation. The deletion test measures faithfulness to the model's own scoring function, but it does not measure human usefulness or agreement with expert rationales.

Ablation scope. The ablation study examines projection type, interference, and fusion weight, but it does not yet test contextual embeddings inside the quantum-inspired projection component.

9 Conclusion

This paper presented an intrinsically explainable quantum-inspired formulation for multi-label NFR classification. Requirements are semantic states, labels are projection directions, and predictions are explained through semantic amplitude contributions rather than post-hoc approximations. The theoretical analysis clarifies exactly when the model behaves like cosine-style centroid ranking and when interference, or row-wise normalization in the positive-projection variant, creates cross-label coupling.

Empirically, the method's strongest role is explanatory and diagnostic. The hybrid quantum–SVM model is competitive with TF-IDF baselines and is the strongest non-embedding model in Macro-F1, but its accuracy advantage is not statistically significant at the current dataset size; pure projection variants are weaker as standalone classifiers. The deletion-based evaluation shows that top contribution terms are substantially more influential than random terms, supporting the faithfulness of the intrinsic explanations. Thus, the paper contributes a transparent multi-label scoring framework and a careful account of when its quantum-inspired geometry helps, not an inflated accuracy claim.

Future work should integrate contextual embeddings, compare with broader BERT/RoBERTa [9], NoRBERT, and ML-kNN [20] baselines, test a second multi-label NFR corpus, and collect human rationale judgments.

Reproducibility

The artifact includes all scripts, data-processing code, reports, and pinned dependencies needed to reproduce the experiments, including an aggregate runner script. For anonymous review, it is available through the anonymous artifact repository.

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